

5 Limitations

We assume that real-world data is a finite set. This reflects the scenario of data-scarce domains. Outside this domain, when the proportion of real-world data in the training corpus does not approach zero as the generation number n increases, LMs may still collapse. However, their output probabilities will not approach arbitrarily closely a term composed solely of the errors introduced in each generation, as stated in our Main Theorem.

Our work does not account for post-processing techniques applied to generated data. It also does not consider commonly used preprocessing methods for filtering data before training new LMs. Future research could address these limitations by relaxing our assumptions to accommodate more scenarios.

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